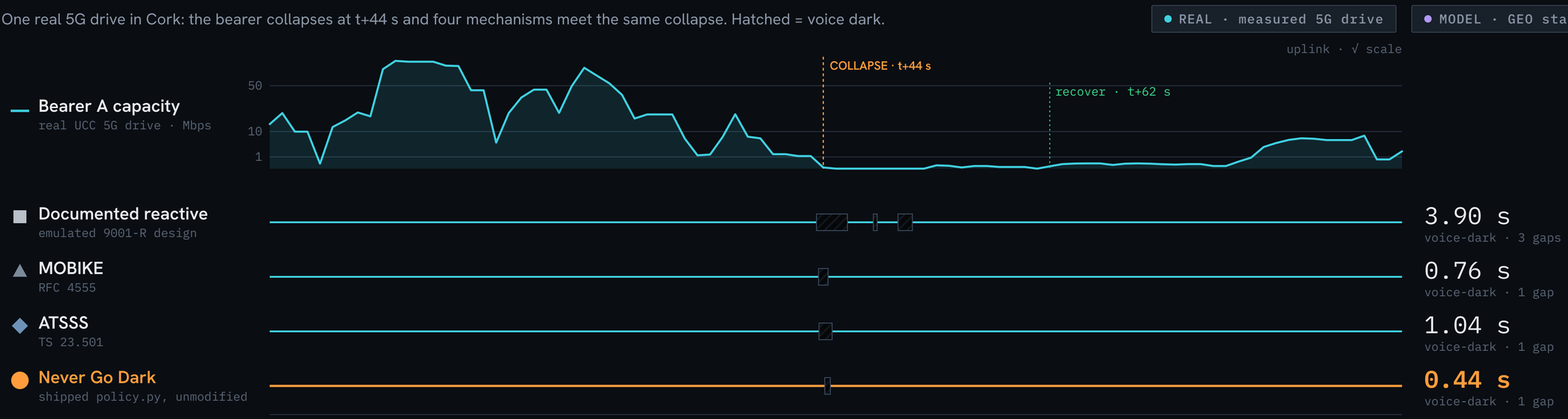
# NEVER GO DARK

## Predictive make-before-break for encrypted tunnels on the move.

EXECUTIVE SUMMARY

Teams on the move lose their encrypted tunnel for seconds when a link drops. Never Go Dark predicts the collapse and finishes the backup handshake before the link fails, leaving the crypto core untouched. It earns the best composite of every existing mechanism we emulated, ATSSS-level continuity at 42% less standby airtime, and is confirmed on real WireGuard. It is a software-only adaptation of the documented KATIM (EDGE Group) 9001-R.

One real 5G drive in Cork: the bearer collapses at t+44 s and four mechanisms meet the same collapse. Hatched = voice dark.

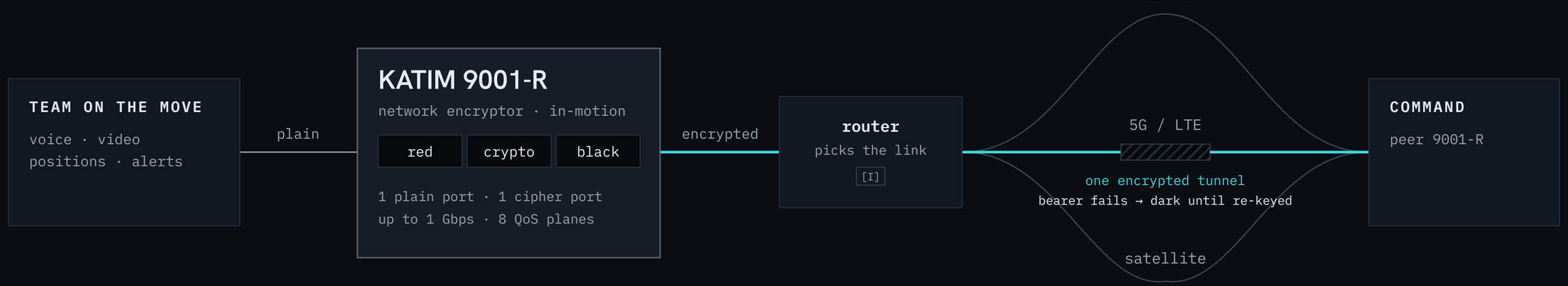



Mohammad Al Ali · Aya Elsayed · Thani Almarzooqi · chart: real UCC 5G uplink (√ scale), single measured prototype run, prototype/logs/result_*_final.json · built only from public 9001-R documentation · not affiliated with EDGE.

# When the bearer dies, the encrypted tunnel goes dark. Every existing fix trades dark gaps for an always-on standby.

## HOW IT WORKS, AND WHERE IT GOES DARK

KATIM (EDGE Group) 9001-R · rugged network encryptor for vehicles and field teams




[D] post-quantum handshake · [D] up to 8 encrypted QoS planes, own keys each · [D] handshake retry every 10 s · [I] one cipher port: the encryptor cannot see the radio

### ■ Documented reactive [D]

emulated 9001-R design

Recovers only after the peer is declared dead, retrying every **10 s**. No standby.

**2.52 s** outage / transition · **6.7 s** standby airtime

### ▲ MOBIKE-style roaming [D]

IETF RFC 4555

Moves the live session in one round trip, but only after dead-peer detection.

**1.23 s** outage / transition · **43.3 s** standby airtime

### ◆ ATSSS active-standby [D]

3GPP TS 23.501 §5.32

Lowest outage of the three, but the standby transmits all the time.

**0.87 s** outage / transition · **98.6 s** standby airtime

**PROBLEM STATEMENT**

An in-motion encryptor that cannot see the radio can only react after the bearer fails: it waits to declare the peer dead, then rebuilds the handshake, so each handover costs seconds of dark voice and stale positions. Today's alternative, a standby that is always on, closes the gap only by transmitting all the time.

# It predicts the collapse from the gateway's own telemetry, and warms the standby before the break.

### Predict, don't react

Warms the standby from risk, seconds before the active link stalls.

### Only the timing changes

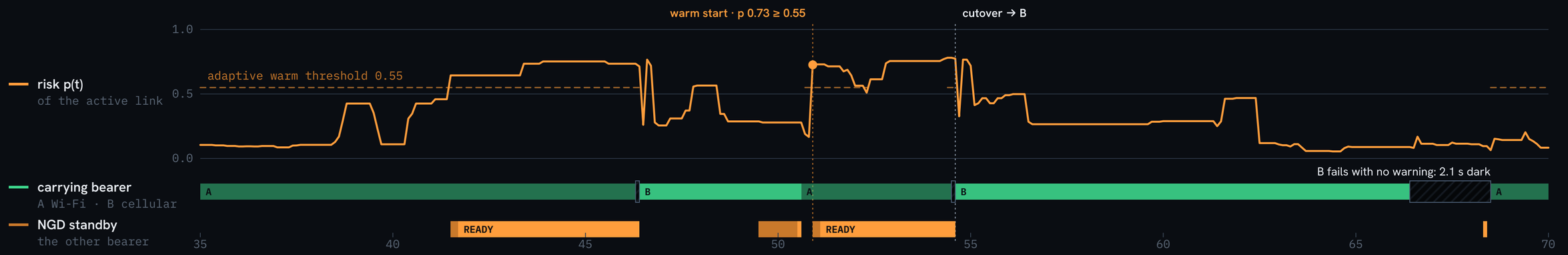
Standard handshakes, fresh keys; the crypto core is untouched.

### Blind to the radio, by design

Uses only the gateway's own encrypted-side telemetry.

COLLAPSE RISK P(T) · REAL HELD-OUT INSTANCE S4:T304_AC · RUN 35-70 S

● REAL | STITCHED · Wi-Fi + UCC 4G LTE




Two predicted collapses: each time, the standby is ready before the active link stalls, so each costs one 0.1 s stall tick. At 66.4 s bearer B fails with no warning and the tunnel is dark for 2.1 s.

## THE PREDICTOR

Inputs are what the gateway already measures on its encrypted side (**keepalive RTT and its trend, loss, staleness, delivered goodput, queue backlog**) and each tunnel's configured bearer class. No modem counters, no router integration. Gradient-boosted trees compiled to plain Python, an **adaptive per-link threshold**, and an early-warm gate for slow satellite handshakes. Only the **timing** of standard handshakes changes.

The prediction is blind to the radio. A pre-registered modem-hint variant **failed on unseen rail** (score −0.123, 6.0 s outage on S5), so the no-radio design shipped.

SHIPPED PREDICTOR · VALIDATION SPLIT (BY TRACE)

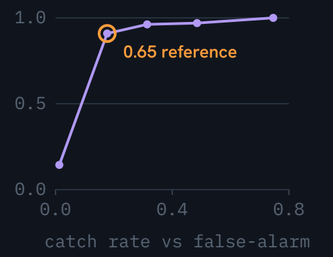
no-modem

**0.832**
ROC-AUC · logistic 0.640

**120/132**
caught (~91%) at 0.65

**4.9 s**
median lead, bounded by the 5 s horizon

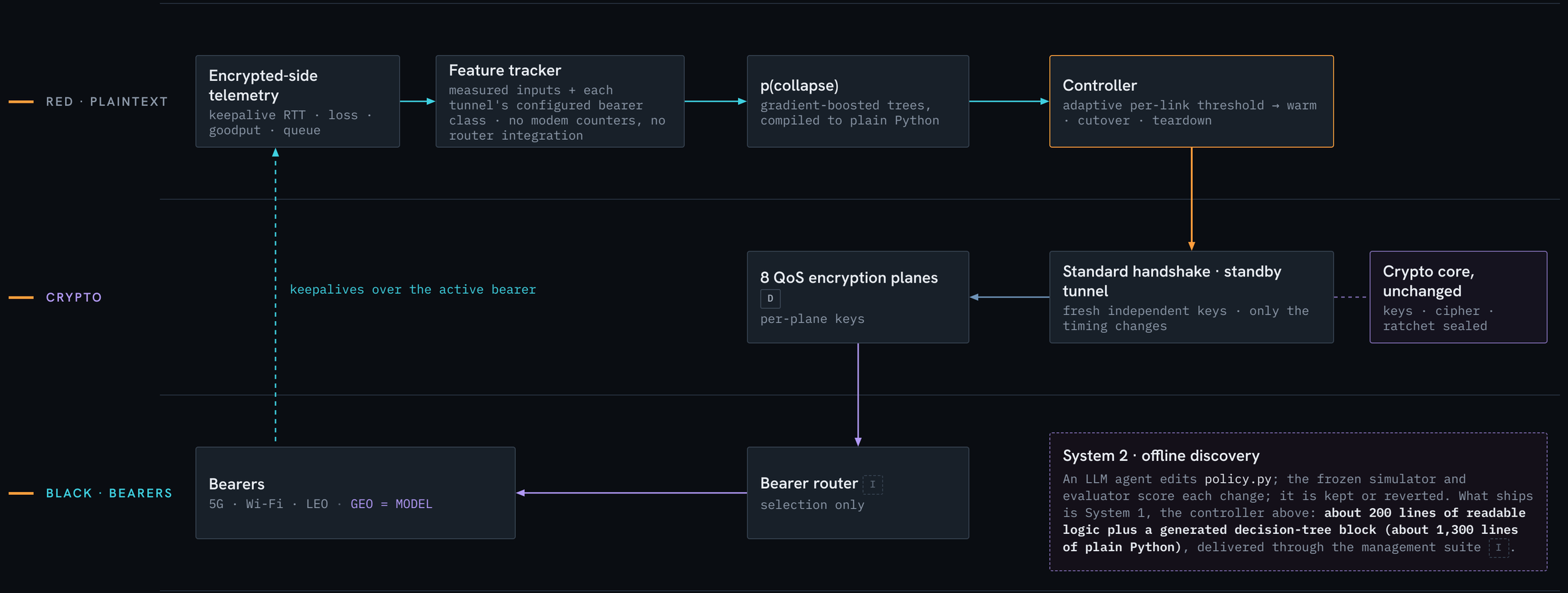
**0.177**
false-alarm tick fraction




N = 132 collapse events over 40,073 validation ticks

p(t), threshold, standby and carrying bearer: held-out instance S4:t304_ac run through the shipped policy (heldout.json, seed 300) · predictor: predictor/report.json, no-modem model, a separate experiment from the outage results.

# A small, auditable policy times standard handshakes. The crypto core never changes.




Documented [D]: 8 QoS-aware encryption planes with per-plane keys; red/crypto/black separation; one active cipher port. Inferred [I]: an external router that only selects the bearer; delivery through the management suite. Policy size measured from policy.py (1,601 lines).

# Seven real datasets, eight scenarios, scored once on fresh seeds the loop never saw.

## 7 MEASURED DATASETS · NO INTERPOLATION, EVER

- UCC 5G · 676 real switches
- UCC 4G LTE
- Lumos5G mmWave · 1,718 blockages
- Starlink on the Road
- Meireles V2I Wi-Fi
- HSR-MPTCP · simultaneous
- Berlin V2X · event truth
- GEO = measurement-parameterised model

## 8 SCENARIOS · HOW EACH STANDBY IS BUILT

| | | | |
|---|---|---|---|
| **S1** REAL<br>LTE↔5G walk<br>single real replay | **S2** STITCHED<br>5G→LEO drive<br>UCC 5G + Starlink LEO, concatenated | **S3** MODEL<br>5G→GEO drive<br>UCC 5G + GEO model | **S4** STITCHED<br>Wi-Fi edge<br>Meireles Wi-Fi + UCC 4G, time-aligned |
| **S5** SIMULTANEOUS<br>HSR rail<br>two carrier subflows, no stitching | **S6** MODEL<br>collective 5G→GEO<br>UCC 5G + GEO model · 15 users | **S7** STITCHED<br>Lumos walk<br>Lumos walk + UCC 4G | **S8** STITCHED<br>Lumos drive<br>Lumos drive + Starlink LEO |

## HOW THE POLICY WAS FOUND, AND AUDITED

1 · LOOP 1

### An LLM agent edits policy.py

The frozen evaluator scores each change on training traces; it is kept only if the score rises.

2 · INDEPENDENT AUDIT

### Over-fitted constants found

Loop 1 had tuned per-scenario constants. We replaced them with an adaptive per-link threshold.

3 · LOOP 2 · FOLD-GUARDED

### No change generalised

7 iterations across trace groups kept no change, so the simpler policy shipped.

### Protocol

100 ms tick · zero-order hold, never interpolated · transitions anchored on real degradation events, with randomised timing · static isolation checker on every policy · split **by trace** · oracle = perfect-foresight ceiling · one run on **seeds 300–319**.

### Score, frozen before the test

```
cost  = 1.0·AoI p95 + 1.0·outage
      + 0.3·overhead + 0.3·airtime − 0.2·video
score = cost(documented) − cost(policy)
      − 10 × gate shortfall
```

AoI, outage and video are scaled by the documented design, overhead and airtime by the always-warm reference, so **the documented design = 0**. The evaluator's SHA-256 (a78f791c…) is recorded in the one-shot file and unchanged.

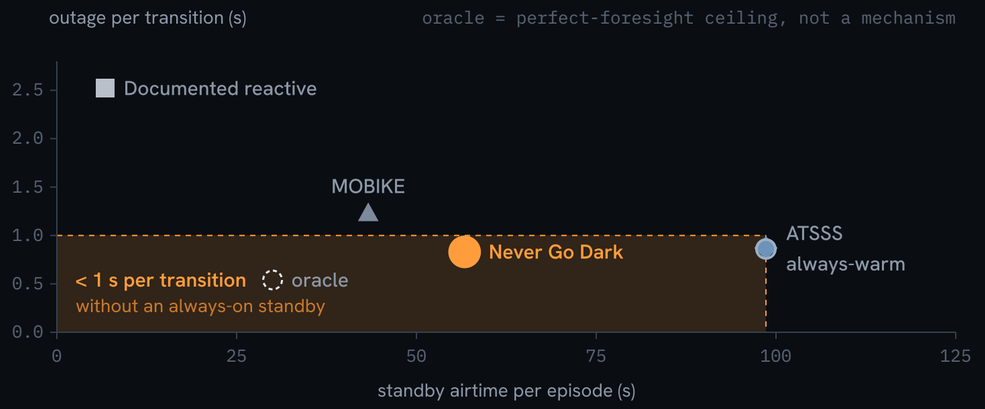
### Disclosure

The held-out traces were evaluated once earlier (seeds 100–119) by an independent auditor on an older policy. That audit led us to remove per-scenario constants. **Nothing was tuned on held-out data.** The final test ran once on fresh seeds 300–319.

Two separate experiments: the outage and airtime results (held-out traces) and the predictor validation (a by-trace split of the training traces) are reported apart. Datasets: SPEC §3, FINDINGS §8 · protocol: SPEC §2, §6, §8 · weights: eval/evaluate.py · scenario construction: heldout.json.

# Best composite of every existing mechanism we emulated, on unseen traces, scored once.

## OUTAGE VS STANDBY AIRTIME · NGD LANDS IN THE EMPTY REGION




**+0.373**

**Best composite score** of every existing mechanism we emulated. Next best: MOBIKE at +0.166.

**0.83 s at 56.7 s**

**ATSSS-level continuity** (0.87 s) with **42% less** standby airtime than ATSSS (98.6 s).

**3× · 1.5×**

**Less outage** than documented reactive and MOBIKE, at more standby airtime than those two (6.7 s and 43.3 s).

| MECHANISM | SCORE | OUTAGE | AIRTIME | STALE P95 | ALERTS | VOICE | VIDEO |
|---|---|---|---|---|---|---|---|
| Documented reactive<br>emulated 9001-R design | −1.269 | 2.52 | 6.7 | 3.88 | 0.952 | 0.912 | 0.779 |
| MOBIKE-style<br>RFC 4555 | +0.166 | 1.23 | 43.3 | 2.26 | 0.986 | 0.962 | 0.803 |
| ATSSS-style<br>TS 23.501 §5.32 | +0.097 | 0.87 | 98.6 | 2.17 | 0.991 | 0.970 | 0.783 |
| Always-warm<br>naive make-before-break | +0.142 | 0.86 | 98.6 | 2.16 | 0.992 | 0.972 | 0.789 |
| **Never Go Dark**<br>shipped policy | **+0.373** | **0.83** | **56.7** | **2.00** | **0.992** | **0.981** | **0.844** |
| Oracle<br>ceiling, not a mechanism | +0.759 | 0.54 | 30.0 | 1.89 | 0.995 | 0.982 | 0.841 |

Seconds: outage per transition; airtime = transmit on the non-primary bearer per episode — a pre-warmed standby, or the active tunnel after a make-before-break roam (so MOBIKE, which pre-warms no standby, still shows airtime once it roams onto B). Rates over feasible messages: alerts on time, voice within 400 ms, video goodput. Replay reproduces outage and gate rates exactly; airtime within one 100 ms tick.

## PER-SCENARIO HONESTY · NEVER GO DARK − ATSSS, PAIRED 95% CI

Clear wins on Wi-Fi edge and rail, one loss on the GEO model, within noise elsewhere. Always less standby airtime than ATSSS.

| S1 REAL | S2 STITCHED | S3 MODEL | S4 STITCHED | S5 SIMUL. | S6 MODEL | S7 STITCHED | S8 STITCHED |
|---|---|---|---|---|---|---|---|
| LTE↔5G walk | 5G→LEO drive | 5G→GEO drive | Wi-Fi edge | HSR rail | collective 5G→GEO | Lumos walk | Lumos drive |
| 0.00 s | +0.07 s | +0.15 s | −0.51 s | −0.89 s | +0.02 s | +0.12 s | +0.01 s |
| outage [0.00, 0.00] | outage [−0.10, +0.23] | outage [+0.05, +0.23] | outage [−1.18, −0.07] | outage [−1.27, −0.55] | outage [−0.17, +0.18] | outage [−0.20, +0.45] | outage [−0.14, +0.16] |
| **identical** | **within noise** | **ATSSS better** | **NGD better** | **NGD better** | **within noise** | **within noise** | **within noise** |
| airtime 0.0 s | airtime −27.8 s | airtime −22.4 s | airtime −96.8 s | airtime −53.8 s | airtime −20.3 s | airtime −74.2 s | airtime −73.1 s |

28 unseen instances · 8 scenarios · 7 real datasets · seeds 300–319, run once. All values: final_heldout_s300-319.json · CIs: paired bootstrap (B = 10,000) over the same (instance, seed) runs, noise_floor.json.

# We ran the shipped policy, unmodified, on real WireGuard tunnels.

CONSOLE · T = 68 S

● REAL · UCC drive

● MODEL · GEO

audio illustrative




The interactive console and the 1:58 film are submitted alongside this document.

REPEATED 60× · 3 SETUPS × 5 RUNS

least-dark in all 3 setups · mean ± sd

**Cork 5G + GEO** ● MODEL

**0.42 ± 0.24 s**

documented 3.60 ± 0.69 s

**Cork 5G + Starlink LEO**

■ STITCHED

**0.50 ± 0.07 s**

documented 2.75 ± 0.01 s

**HSR rail** ◉ REAL SIMULTANEOUS

**0.00 ± 0.00 s**

documented 6.10 ± 0.75 s

ONE MEASURED RUN · A = REAL 5G · B = GEO MODEL

| | | |
|---|---|---|
| ■ Documented reactive | 3.90 s | voice-dark · standby 2.4 s · ≤400 ms 0.910 |
| ▲ MOBIKE | 0.76 s | voice-dark · standby 0.0 s · ≤400 ms 0.966 |
| ◆ ATSSS | 1.04 s | voice-dark · standby 89.5 s · ≤400 ms 0.982 |
| ● Never Go Dark | 0.44 s | voice-dark · standby 63.2 s · ≤400 ms 0.992 |

The shipped `policy.py` runs read-only on the real tunnels and drives real warm, handshake and cutover operations. It had the smallest dark gap of the four: **8.9×** less than reactive, **2.4×** less than ATSSS and **1.7×** less than MOBIKE, with 29% less standby airtime than ATSSS (this run; –31% over 5 runs).

single measured run · prototype/logs/result_*_final.json

Real WireGuard in Docker · real UCC 5G drive (Cork, 27 Feb 2020) collapsing at ~43.5 s · bearer B = GEO model · loss pattern measured, audio illustrative · repeats: prototype-repeats.json (6 runs set aside, pre-registered rule) · the simulator is the statistical evidence.

# A software rollout in three phases. No new hardware, no change to the keys.

PHASE 1

## Shadow

The policy runs and logs p and the actions it would take. No actuation.

Advance when the field catch rate and false-warm rate match the validation operating point: ~91% caught at a 0.177 false-alarm tick fraction.

PHASE 2

## Advisory

Predicted collapses are shown to operators in the management console [1].

Operators see the lead time and the standby's readiness; the policy still does not act on its own.

PHASE 3

## Active

Per-link warm, cutover and teardown, under a per-profile airtime and emission budget.

The budget caps standby airtime per bearer, so emissions stay inside each mission profile.

### WHAT SHIPS

No modem or router integration: the only bearer fact it uses is each tunnel's configured class. Keys and crypto are untouched. The policy is **about 200 lines of readable logic plus a generated decision-tree block (about 1,300 lines of plain Python)**, delivered as a policy update through the management suite.

### LIMITS WE STATE

An emulation built only from public documents. GEO is a model, not a measured link. The prototype is 60 real-tunnel runs across 3 bearer setups. Always-on wins on the GEO model. The oracle ceiling (0.54 s) shows headroom for better prediction.

**Disclosure**

Nothing was tuned on held-out data; the final test ran once on fresh seeds 300–319.

**Fit with the 9001-R**

Software only. No new hardware, no new radios, no change to keys or cipher. It uses the documented QoS planes and handshake as they are.

PROPOSED FIRST STEP

Shadow mode on vehicles that already carry 9001-R units, logging the predicted risk against real handovers, with no change to their behaviour.

Operating point: predictor/report.json (no-modem) · oracle ceiling: final_heldout_s300-319.json · policy size measured from policy.py · baselines are a faithful emulation of the documented 9001-R design.

# Continuity for teams on the move, at less airtime, with QoS guidance any encryptor can reuse.

## EVIDENCE · ONE SHARED QUEUE VS THE DOCUMENTED 8 PLANES




Merge the emulated 9001-R's planes into one shared queue and voice within 400 ms falls from **0.89 to 0.42**; alerts on time fall from **0.91 to 0.73**. So keep per-class planes.

## INTENDED IMPACT

**01 Continuity for moving field teams and command nodes**

Voice and positions stay up through bearer changes: `0.83 s` outage per transition vs `2.52 s` documented.

**02 Lower emissions and airtime**

`56.7 s vs 98.6 s` of standby airtime against ATSSS, at the same continuity: a smaller emission signature.

**03 Reusable QoS guidance**

Six measured rules, below. The headline: keep per-class planes (one queue: voice `0.89 → 0.42`).

## QOS / QOE GUIDANCE · FROM THE MEASUREMENTS

- G1 Keep per-class encryption planes · one queue: voice 0.42, alerts 0.73
- G2 Never discard in-flight data under staleness · dropping it thrashed freshness to 17–20 s p95
- G3 Warm on predicted risk, not observed loss · median lead 4.9 s, 120 of 132 caught
- G4 Warm satellite standbys early · a GEO handshake takes ~1.2–1.8 s at 600 ms RTT
- G5 Report airtime next to outage · 56.7 s vs 98.6 s at 0.83 vs 0.87 s outage
- G6 Hold the voice targets (TS 22.179, G.114) · within 400 ms: 0.912 documented, 0.981 Never Go Dark

### HOW WE WOULD DEMONSTRATE IF SHORTLISTED

At Demo Days we would show the interactive continuity console with its live mode driving the real WireGuard prototype, a live run of the shipped policy, and the 1:58 film.

Reduced planes: planes_experiment.json (emulated documented design; 8, 2 and 1 planes) · thrashing run: planes_experiment_v1_thrashing.json · lead and catch rate: predictor/report.json · airtime, outage and voice: final_heldout_s300-319.json · GEO handshake: SPEC §4.3.

# Who built it, what we assumed, and what you can check.

## TEAM

| | |
|---|---|
| Mohammad Al Ali | coder |
| Aya Elsayed | researcher · reviewer / coder |
| Thani Almarzooqi | coder |

## SUPPORTING MATERIALS

Interactive continuity console · replay of every result, plus live mode on the WireGuard prototype

The 1:58 film · the problem, one measured drive, the result

The evidence pack · result files, claims register, simulator and evaluator

All built from public documents · not affiliated with EDGE.

## ASSUMPTIONS, STATED PLAINLY

**A1** GEO is a measurement-parameterised model, not a live link.

**A2** Stitched scenarios are independently measured and only time-aligned.

**A3** The KATIM internals we build on are derived from public documents and tagged documented `D` or inferred `I`.

**A4** One measured drive and the WireGuard prototype illustrate the design; the 28 held-out instances are the statistical evidence.

**Disclosure**

Nothing was tuned on held-out data; the final test ran once on fresh seeds 300–319.

## AI DECLARATION

AI assisted tools aided in the research and design of this project.

# NEVER GO DARK

ATP2026 · EDGE track · predictive make-before-break for encrypted tunnels